



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Assignment 2 – FALL 2025

Course Title:	Microprocessor System and Interfacing			Course Code:	CPE 342	Credit Hours:	4(3,1)
Course Instructor:	Dr. Abbas Javed, Mr. Moazzam Ali Sahi			Program Name:	BCE		
Semester:	5 th	Batch:	FA23	Section:	A & B	Date Given:	October 22, 2025
Submission Date:	October 28, 2025			Maximum Marks:	30		
Name:				Registration Number:			

Important Instructions / Guidelines:

- **Do your own work, PLAGARISM will be graded as ZERO.**
- **No late submission.**
- Provide full screen snapshot of results, clearly indicating time and date from menu.
- Only handwritten assignments will be graded
- Use this as the title page of your assignment.

Question 1: **[CLO2-PLO3-C5]** **[3+10+2 Marks]**

A **32 KB** memory chip is used to build a memory system with the starting address **30000H**. During testing, it was found that a block of memory locations (**31400H–317FFH**) within the chip is damaged. You are required to analyze the memory map and suggest a replacement block for the damaged section.

Tasks:

1. **Prepare** the memory map showing:
 - The **1 MB** address space.
 - The location of the **32 KB** memory chip.
 - The damaged block within that chip.
 - Determine the **ending address** of the memory chip.
2. **Design** the **circuit diagram** for interfacing the 32 KB memory and for replacing the damaged part with a replacement memory chip.
3. **Summarize**, with address calculations, how the system will now skip the damaged section and correctly access the new replacement memory block.

Question 2: **[CLO2-PLO3-C5]** **[10 +2+3] Marks**

You are required to **interface a 200 KB SRAM** with an **8086 microprocessor**. The available memory chips have capacities of **16 KB, 8 KB, or 4 KB**. Design a suitable interfacing circuit using an appropriate starting address.

Tasks:

1. **Create** a fully labeled schematic diagram showing the complete memory interfacing with the 8086 CPU.
2. **Prepare** the memory map for each memory chip used in your design, clearly indicating the starting and ending addresses of each memory block.
3. **Summarize** your design approach, highlighting how it ensures efficient use of resources and supports scalability for future expansion.

End of Assignment